z-Scores & Linear Transformations of Data

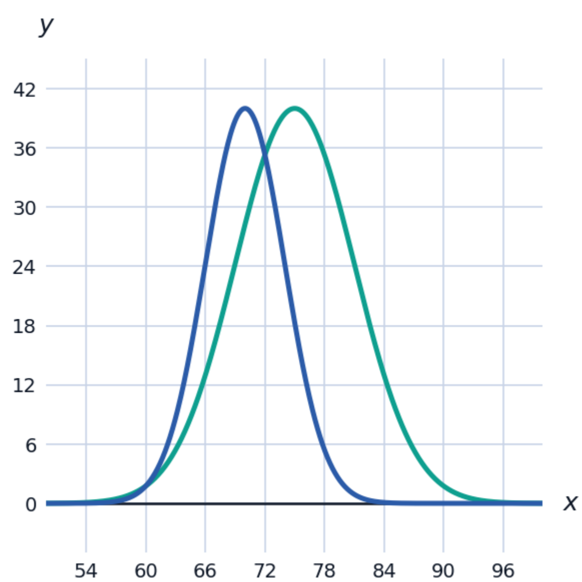
Computing z-scores

1 A test has mean 70 and standard deviation 5. Find the z-score for a student who scored 82.

2 Explain in words what a z-score of -1.5 means, in the context of a data value's position relative to its distribution.

Comparing values across different distributions

3 On Test A (mean 75, standard deviation 6), a student scores 87. On Test B (mean 70, standard deviation 4), a different student scores 77. Compare the two students' relative performance using z-scores.



Effect of adding a constant to every data value

4 A data set has mean 50 and standard deviation 8. If 10 is added to every value, find the new mean and new standard deviation.

5 Explain, without doing a full calculation, why adding a constant to every data value does not change the standard deviation.

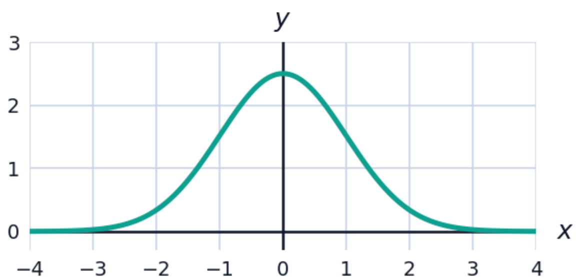
Effect of multiplying every data value by a constant

6 A data set has mean 20 and standard deviation 3. If every value is multiplied by 4, find the new mean and new standard deviation.

7 A distance measured in miles has mean 10 and standard deviation 2. Convert both to kilometers (multiply by 1.609), and find the new mean and standard deviation, to two decimal places.

Visualizing a standardized (z-score) distribution

8 The graph shows the standard normal distribution, which has mean 0 and standard deviation 1 — the distribution that results after standardizing any normal data set using z-scores.



a State the mean and standard deviation shown by this curve.

b Explain why standardizing a data set using $z = \dfrac{x-\mu}{\sigma}$ always produces a distribution with mean 0 and standard deviation 1, regardless of the original data's mean and standard deviation.

Applying linear transformations in context

9 A factory's daily output (in units) has mean 500 and standard deviation 30. Management adds a fixed bonus batch of 50 units to every day's count, then reports output in dozens (divide by 12). Find the new mean and standard deviation, to two decimal places.

Finding a raw value from a z-score

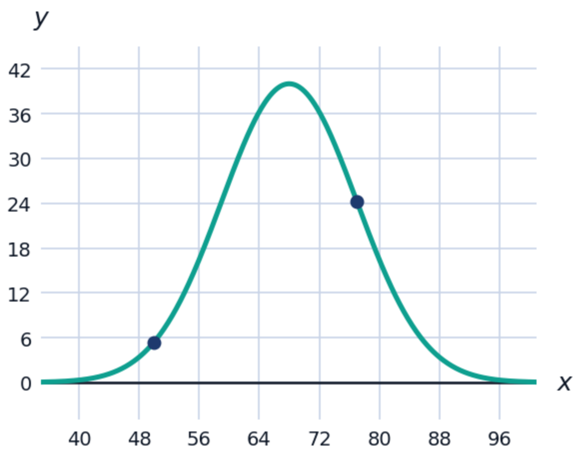
10 A distribution has mean $\mu = 64$ and standard deviation $\sigma = 5$. Find the data value corresponding to a z-score of $z = -1.8$.

11 A distribution has mean $\mu = 200$ and standard deviation $\sigma = 15$. If a data value has $z = 2.2$, find the data value.

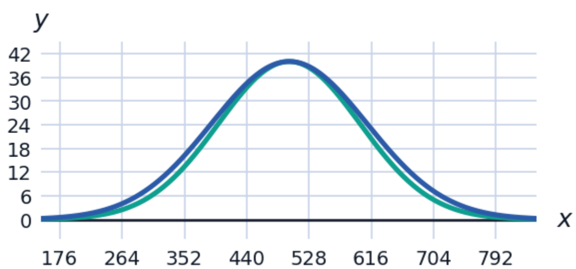
Combined shift-and-scale transformations

12 A teacher curves test scores (mean 65, standard deviation 10) by first adding 5 points to every score, then multiplying the result by 1.1. Find the new mean and new standard deviation, to two decimal places.

13 A weight distribution has mean $\mu = 68$ kg and standard deviation $\sigma = 9$ kg. Find the z-scores for weights of 77 kg and 50 kg.



14 A national math exam has mean 500 and standard deviation 100, while a national verbal exam has mean 500 and standard deviation 110. A student scores 640 on the math exam and 625 on the verbal exam. Even though the verbal raw score is lower, use z-scores to determine which exam the student performed better on relative to other test-takers, and explain your reasoning.



15 A shipping company records package weights with mean 18 kg and standard deviation 2.5 kg. The company then converts every measurement from kilograms to pounds by multiplying by 2.20462, and separately subtracts 1 kg from every package to account for packaging weight before converting. Describe, step by step, how each operation (the subtraction, then the multiplication) affects the mean and standard deviation.